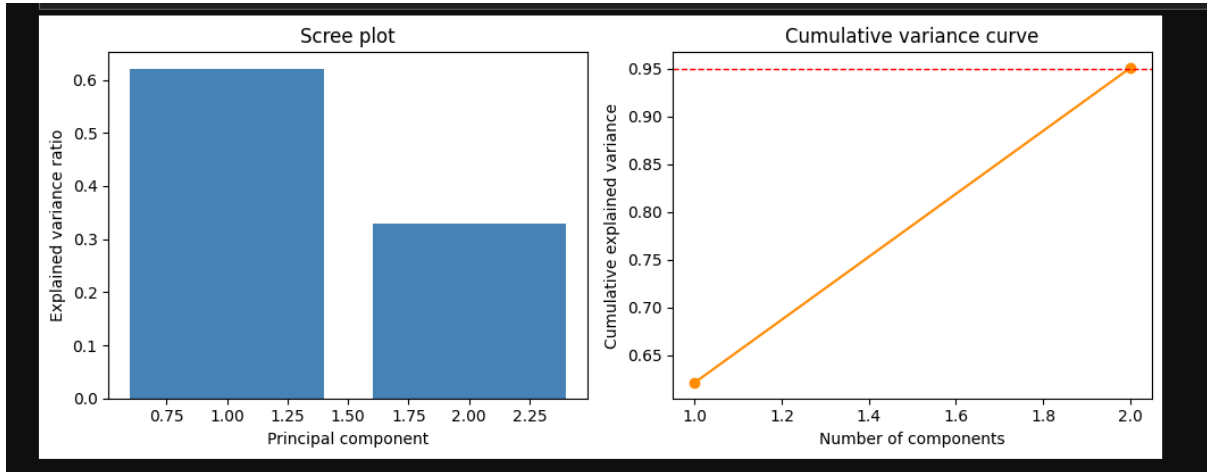


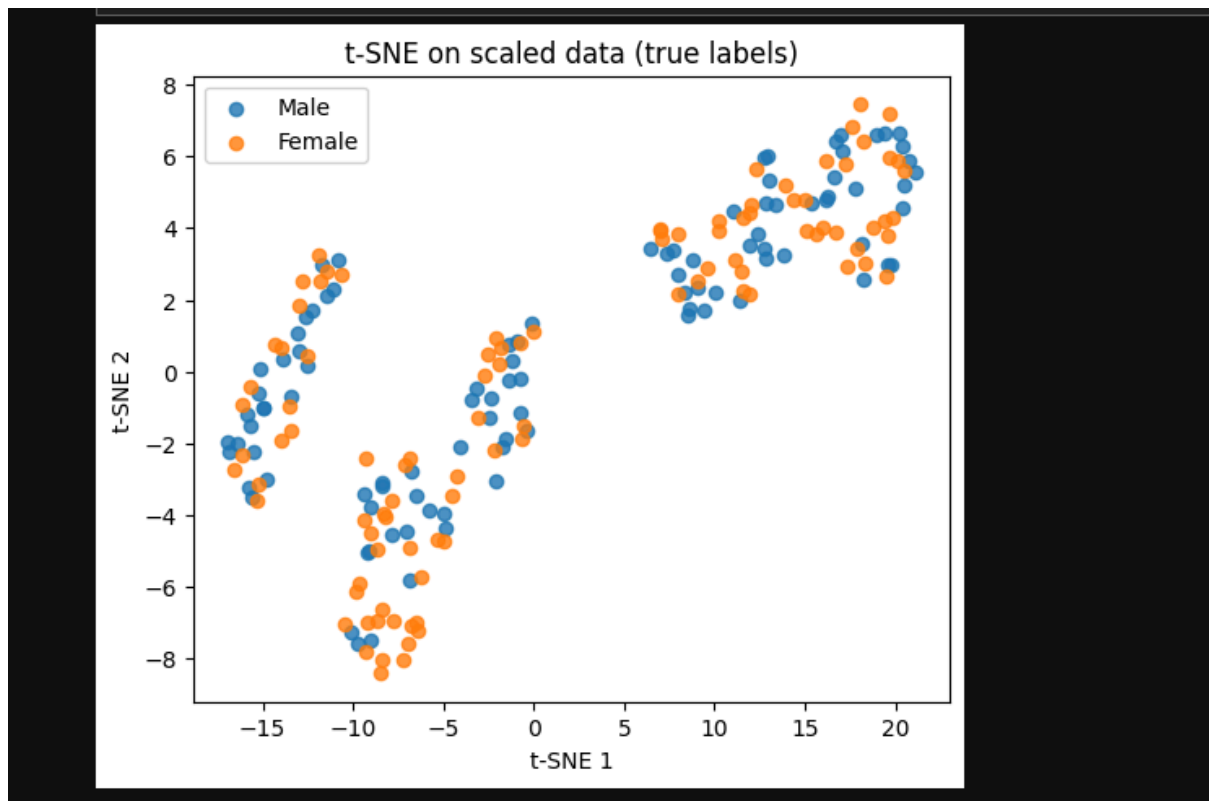
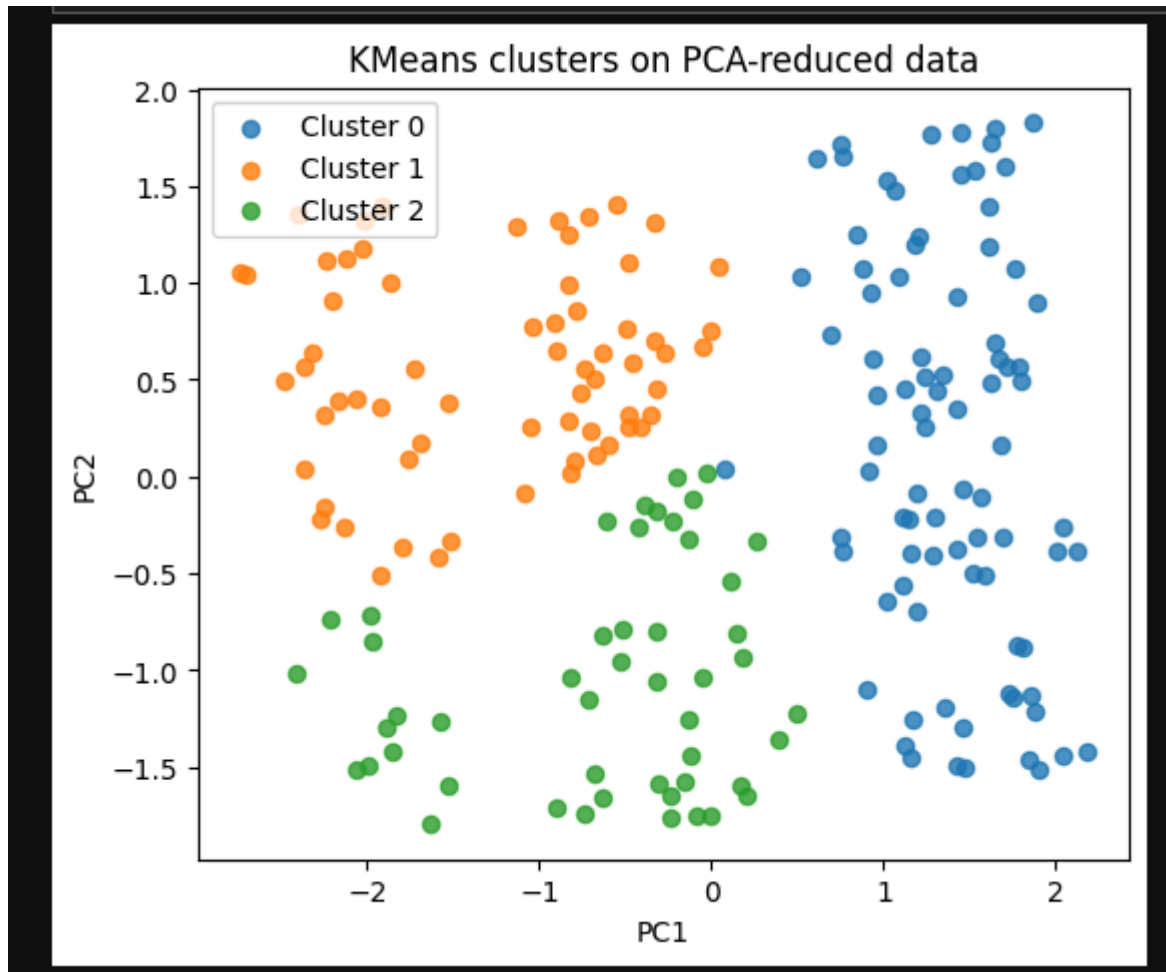
OUTPUTS:

Q1.

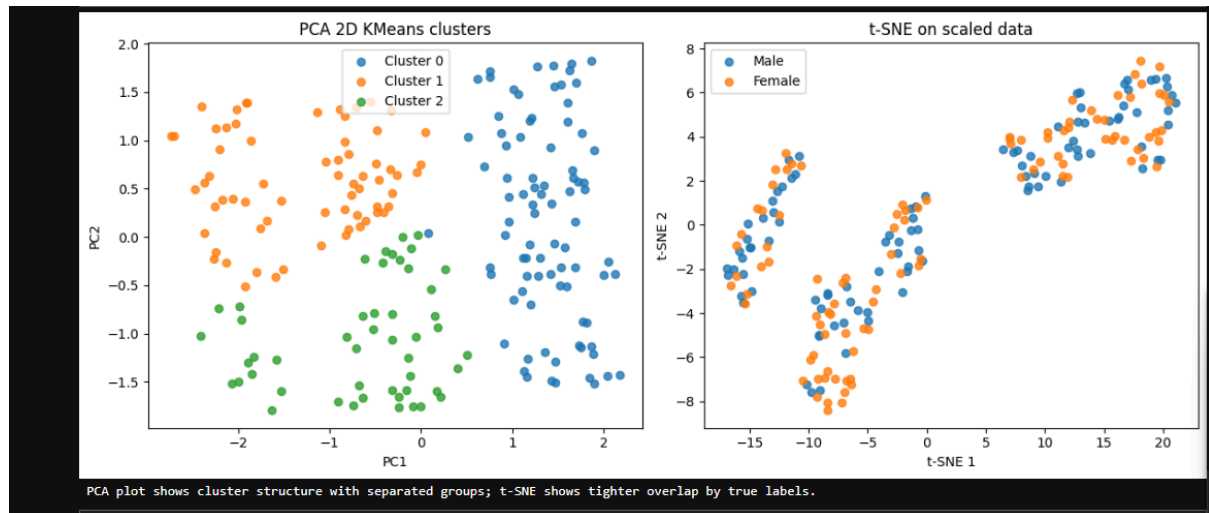
1A.



1B.

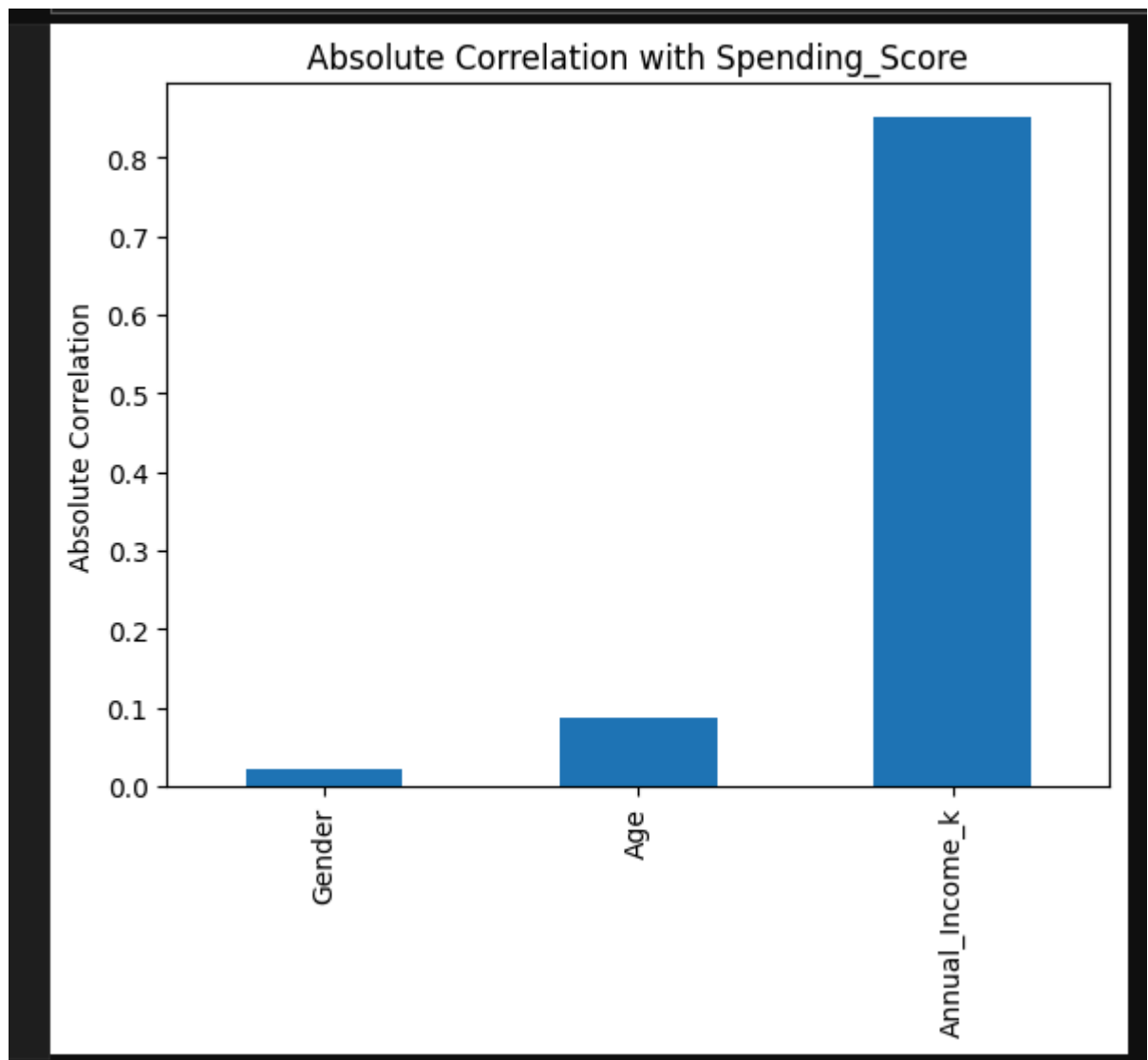


1C.

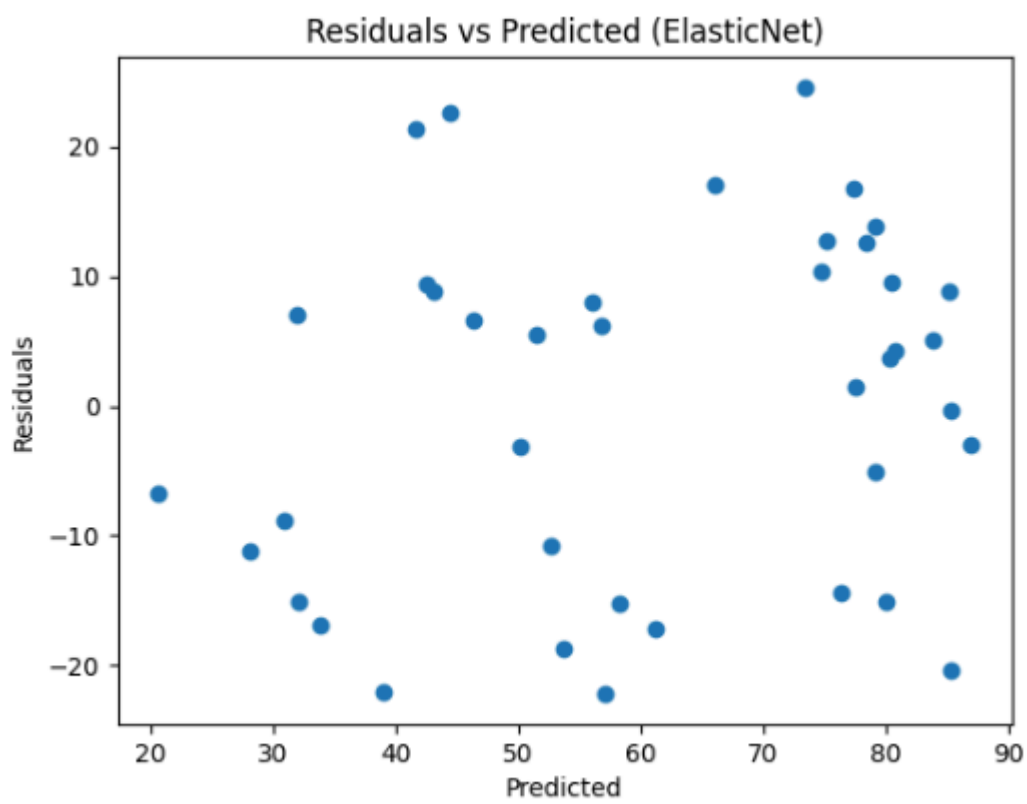
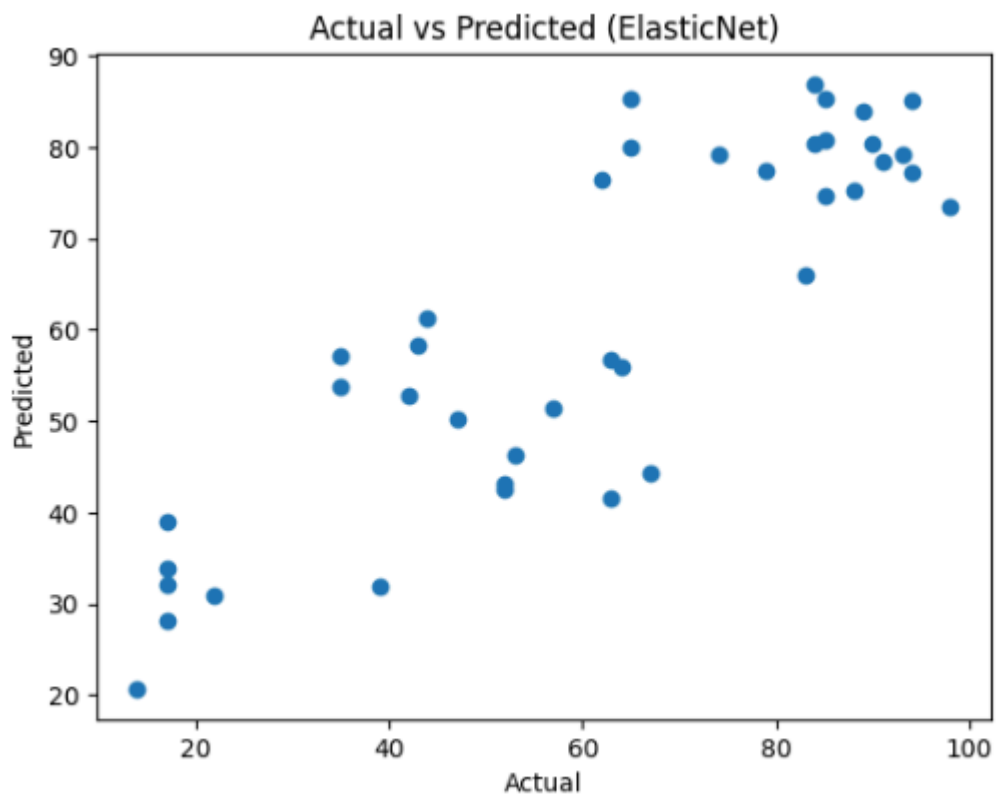


Q2.

2A.

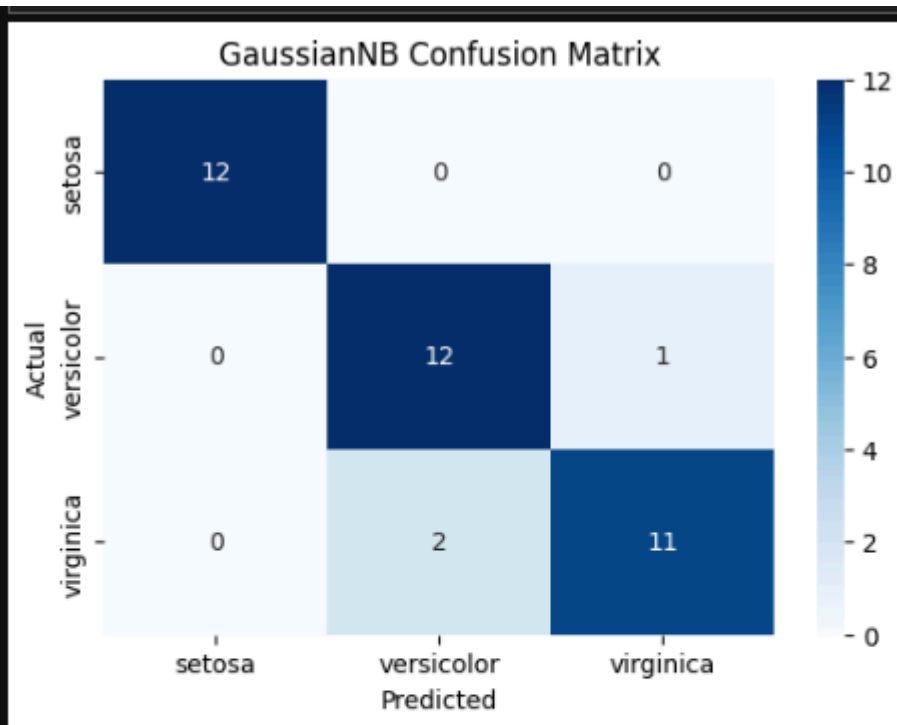


2C



Q3.

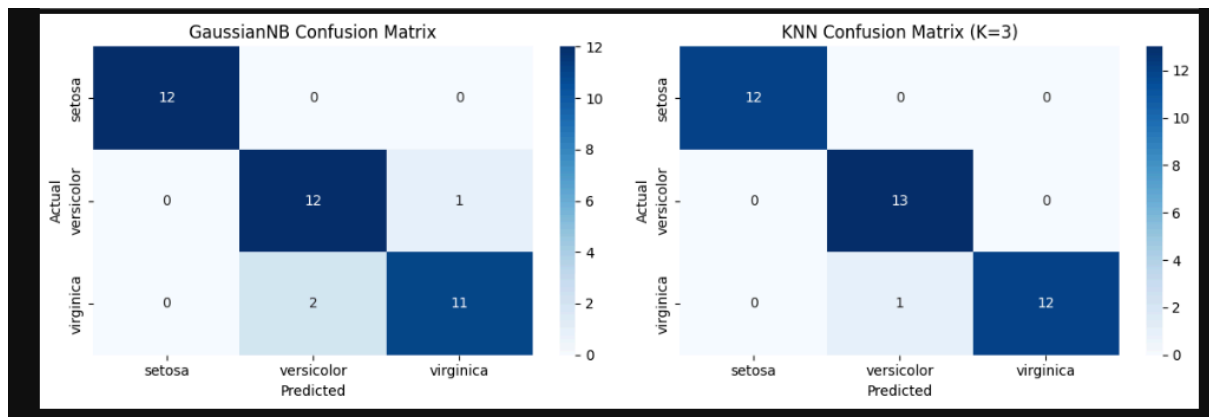
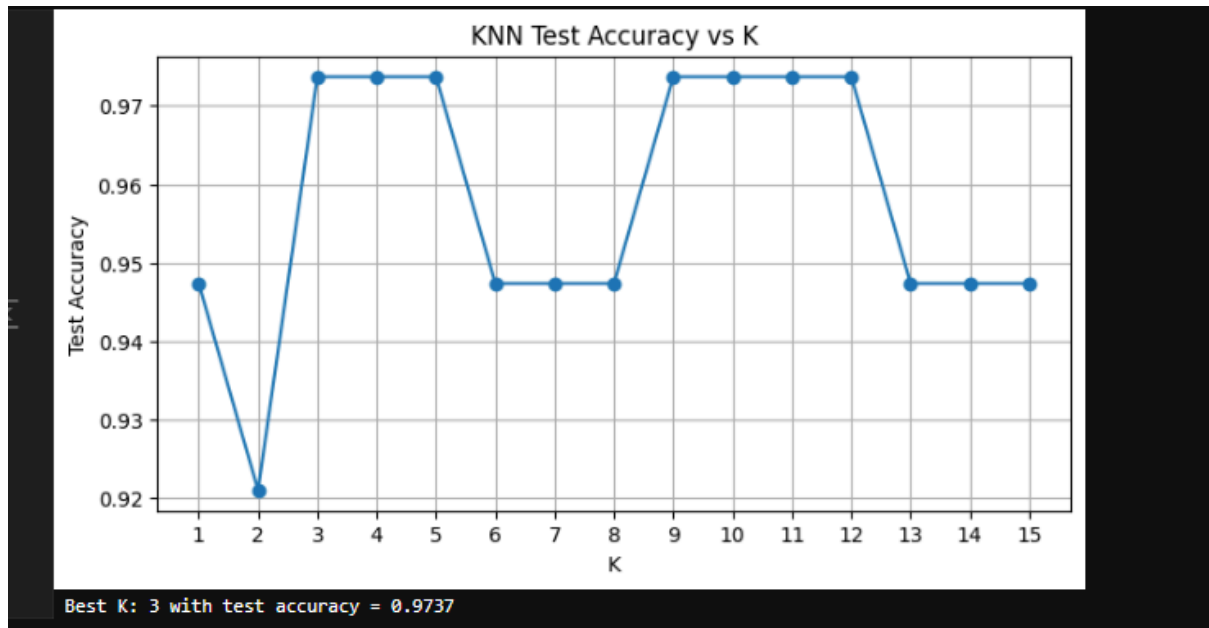
3A



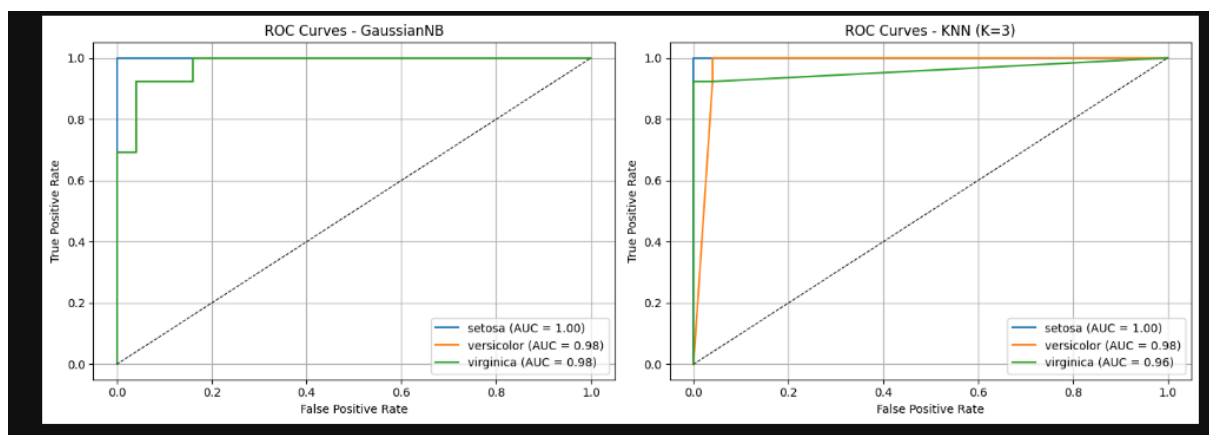
GaussianNB Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	12
versicolor	0.86	0.92	0.89	13
virginica	0.92	0.85	0.88	13
accuracy			0.92	38
macro avg	0.92	0.92	0.92	38
weighted avg	0.92	0.92	0.92	38

3B.



3C.

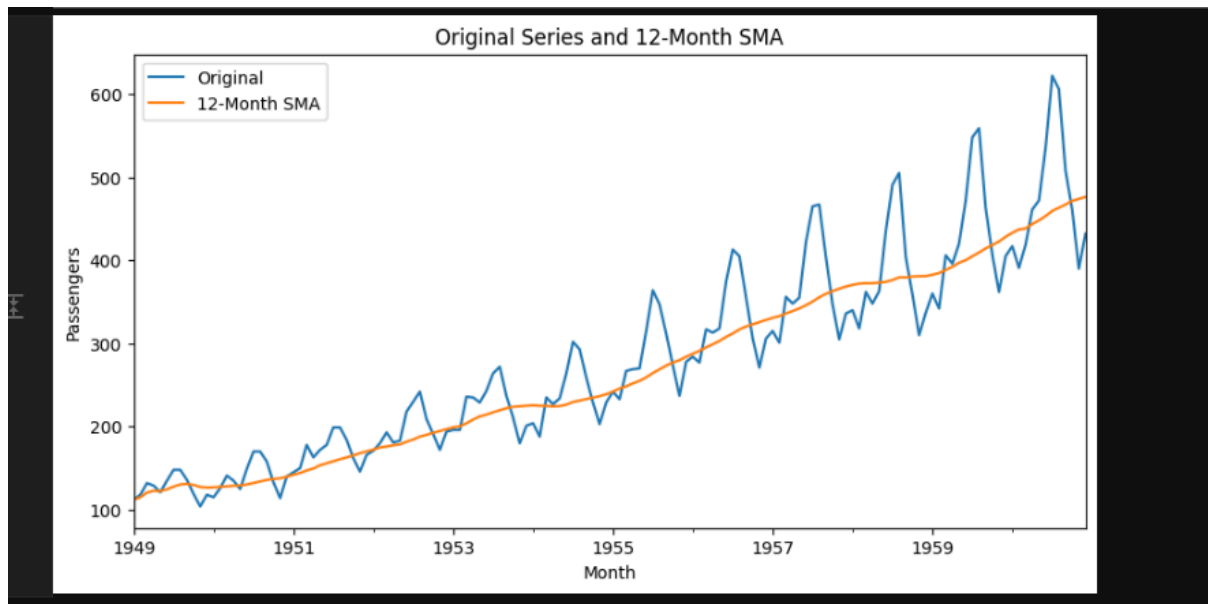
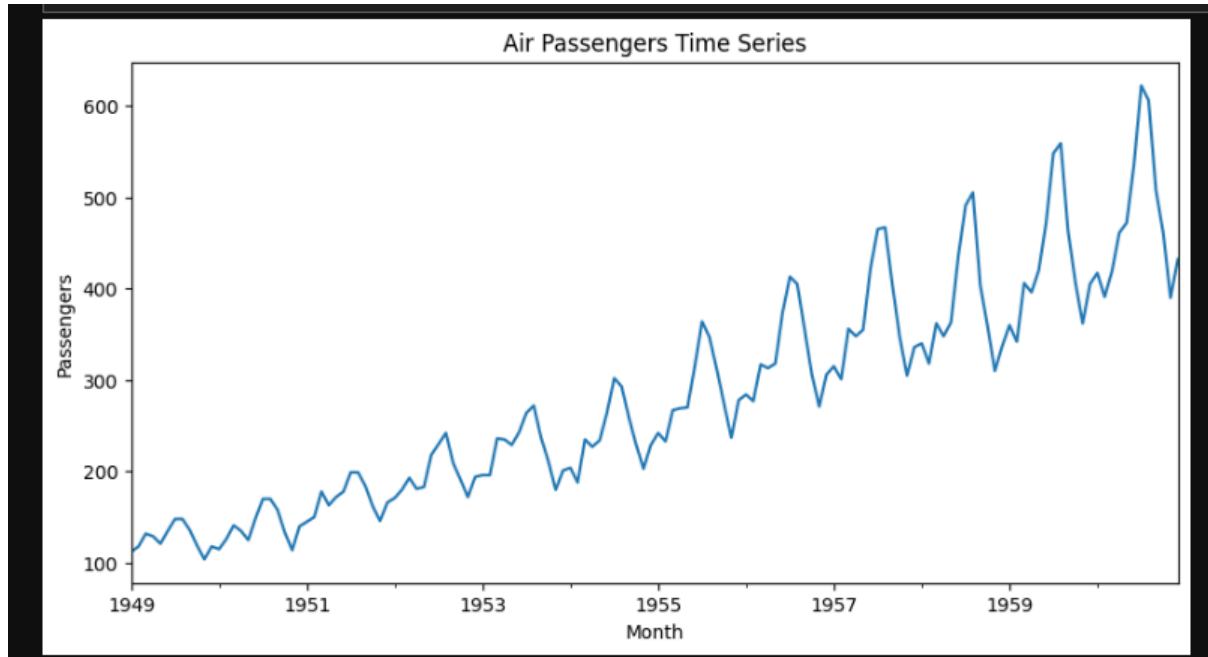


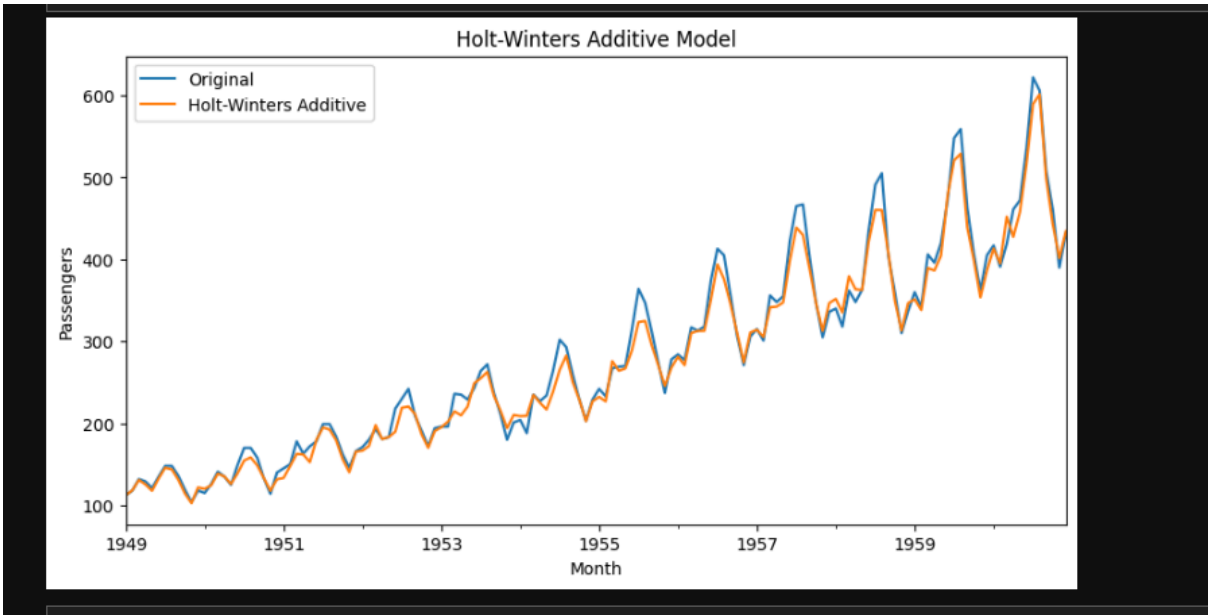
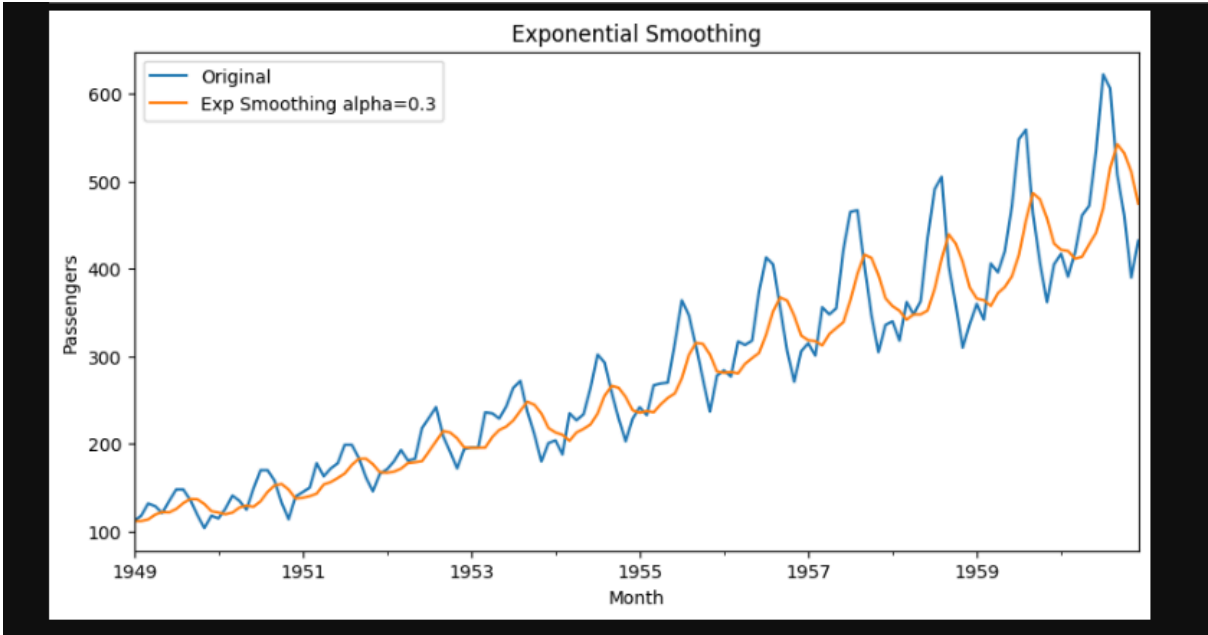
	Model	Accuracy	Precision (macro)	Recall (macro)	F1 Score (macro)
0	GaussianNB	0.921053	0.924603	0.923077	0.922963
1	KNN (K=3)	0.973684	0.976190	0.974359	0.974321

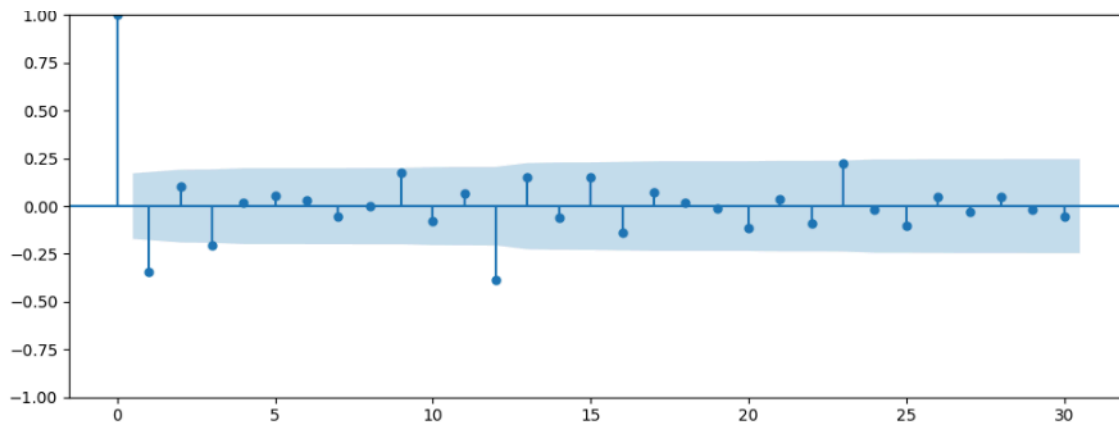
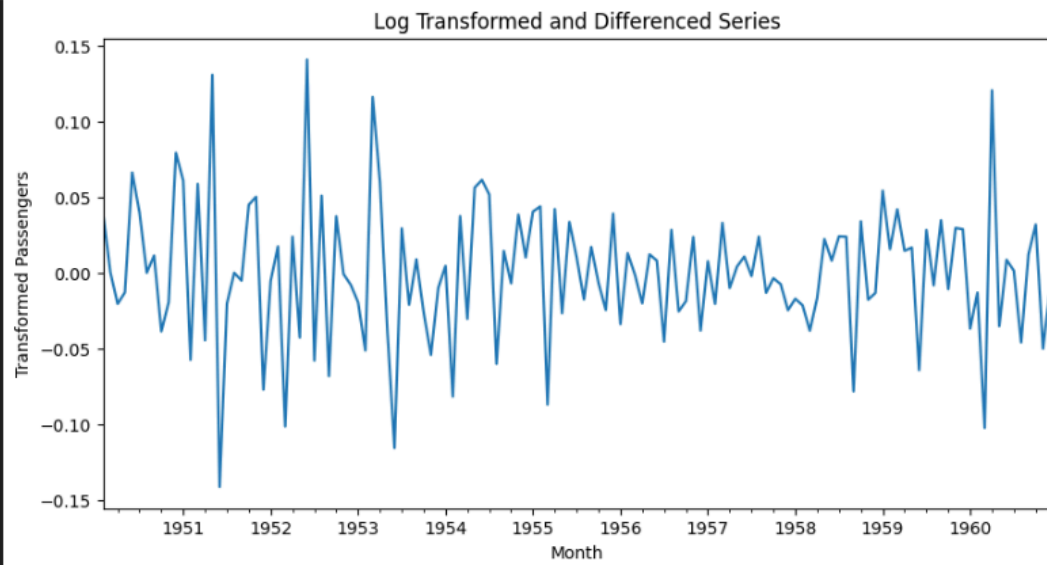
Conclusion:

Based on macro-averaged performance, KNN (K=3) is better suited for this multiclass iris classification task. This conclusion uses accuracy, precision, recall, and F1 score to compare both classifiers fairly.

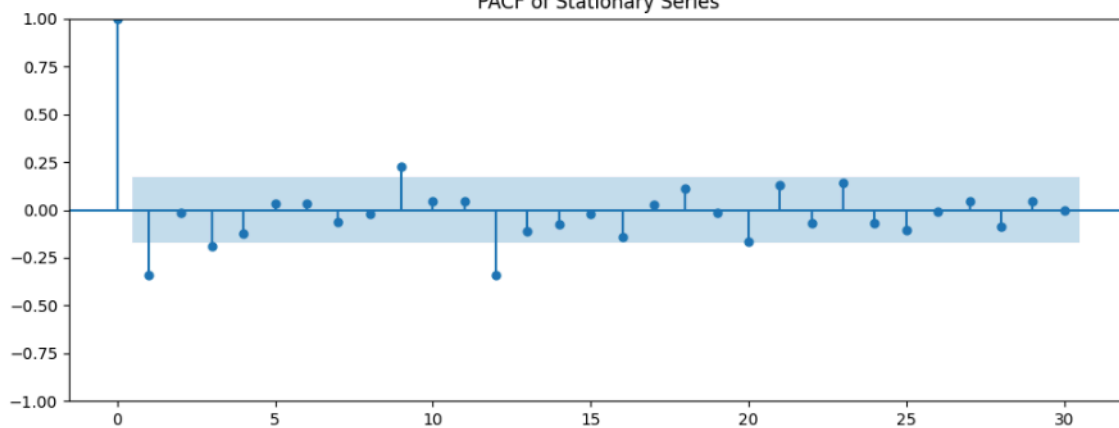
Q4.







PACF of Stationary Series



Appropriate p and q values can be inferred from PACF and ACF cuts after differencing.

```
SARIMAX Results
=====
Dep. Variable:      Passengers      No. Observations:      144
Model:              SARIMAX(1, 1, 1)x(1, 1, 1, 12)      Log Likelihood      -456.103
Date:              Mon, 04 May 2026      AIC      922.205
Time:              13:33:03      BIC      936.016
Sample:            01-01-1949      HQIC      927.812
                  - 12-01-1960
Covariance Type:    opg
=====
              coef      std err      z      P>|z|      [0.025      0.975]
-----
ar.L1         -0.2298      0.401     -0.573     0.567     -1.016      0.557
ma.L1         -0.0987      0.374     -0.264     0.792     -0.832      0.634
ar.S.L12      -0.5460      0.299     -1.825     0.068     -1.133      0.041
ma.S.L12       0.3959      0.352      1.125     0.261     -0.294      1.086
sigma2        140.2945     17.997      7.795     0.000     105.020     175.569
=====
Ljung-Box (L1) (Q):      0.00      Jarque-Bera (JB):      5.42
Prob(Q):                0.95      Prob(JB):      0.07
Heteroskedasticity (H):  2.51      Skew:      0.12
Prob(H) (two-sided):    0.01      Kurtosis:      4.03
=====

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

